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论文题目： 基于改进卷积神经网络的音乐流派分类系统设计与实现

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摘要

音乐流派分类是音乐信息检索（MIR）领域的基础任务，在音乐推荐系统、数字图书馆管理和音频内容分析中有着广泛的应用。传统方法主要依赖手工设计的声学特征与浅层机器学习分类器的结合，难以捕捉音乐中复杂的层次化模式。近年来，深度学习通过直接从原始音频数据中自动学习判别性特征展现出了更优越的性能。

本文对改进的卷积神经网络（CNN）架构在音乐流派分类中的应用展开了全面研究。系统采用 Mel 声谱图作为音频信号的时频表示，并将其作为深层 CNN 模型的输入。本文所提出的架构在标准 CNN 的基础上进行了多项改进，包括引入批归一化以稳定和加速网络训练，构建渐进式卷积块来实现层次化特征学习，采用全局平均池化以大幅减少参数量并降低过拟合风险，同时结合 Dropout 和 L_2 权重正则化手段以强化泛化能力。

针对广泛使用的 GTZAN 数据集标注样本有限的挑战，本研究在频域内实施了基于时间与频率掩蔽的数据增强策略。实验结果表明，该改进的 CNN 模型在 GTZAN 测试集上实现了 88% 的分类准确率，能够有效区分十种不同的音乐流派。最后，本文通过三组典型流派的谱图样本预测结果进行了详细的误差行为分析，证明了该系统具备极高的实用价值与工程落地可行性。

关键词：音乐流派分类，卷积神经网络，深度学习，Mel 声谱图，数据增强，GTZAN 数据集

Abstract

Music genre classification is a fundamental task in music information retrieval (MIR) with wide-ranging applications in music recommendation systems, digital library organization, and audio content analysis. While traditional methods often relied on handcrafted acoustic features combined with shallow machine learning classifiers, recent advances in deep learning have demonstrated superior performance by automatically learning discriminative representations from raw audio data.

This paper presents a comprehensive study on the application of an improved convolutional neural network (CNN) architecture for music genre classification. Our approach utilizes Mel-spectrograms as time-frequency representations of audio signals, which serve as input to a carefully designed deep CNN model. The proposed architecture incorporates several enhancements over standard CNNs, including batch normalization for stabilized training, progressive convolutional blocks for hierarchical feature learning, global average pooling to reduce overfitting, and strong regularization through dropout and L2 regularization.

To address the challenge of limited labeled data in the widely-used GTZAN dataset, we implement spectral domain data augmentation strategies involving time and frequency masking. Experimental results demonstrate that our improved CNN model achieves an accuracy of 88% on the GTZAN dataset, effectively distinguishing between ten musical genres. Detailed analysis through sample predictions provides insights into the model's performance characteristics and learning behavior.

Key Words: Music Genre Classification, Convolutional Neural Network, Mel-Spectrogram, Data Augmentation, GTZAN Dataset

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专用术语注释表

Nomenclature:

Symbol	Descripti
X	Original r
X_{norm}	Normaliz
μ	Spatial st
σ	Spatial st
	tensity
n_{fft}	Number
	Transform
λ	Weight de

List of Abbreviations:

Abbreviation	Full Eng
CNN	Convolut
MIR	Music Inf
STFT	Short-Tim
dB	Decibel
ReLU	Rectified
GAP	Global A
MFCC	Mel-Freq

第一章 Introduction

1.1 Background and Motivation

The exponential growth of digital music libraries and streaming platforms has created an unprecedented demand for intelligent music organization and recommendation systems. Within this context, automatic music genre classification has emerged as a fundamental research problem in the field of Music Information Retrieval (MIR). Genre serves as a primary organizational schema for music collections, influencing discovery, recommendation, and curation processes across commercial platforms and personal libraries alike.

Traditional approaches to music genre classification predominantly relied on handcrafted acoustic features derived from digital signal processing techniques. These features typically included temporal characteristics, spectral features, and Mel-frequency cepstral coefficients (MFCCs). These engineered features were then fed into conventional machine learning classifiers such as support vector machines (SVMs) or k-nearest neighbors (k-NN). While these methods achieved moderate success, they faced inherent limitations in capturing the complex, hierarchical patterns that characterize musical genres, often requiring extensive domain expertise for feature design and selection.

The advent of deep learning has revolutionized the field of audio classification, offering the potential to learn discriminative representations directly from raw or minimally processed audio data. Convolutional Neural Networks (CNNs), in particular, have demonstrated remarkable success by treating time-frequency representations (such as spectrograms) as two-dimensional “images.” This research aims to overcome the limitations of traditional handcrafted features by introducing an improved CNN architecture, achieving competitive classification performance on benchmark datasets.

1.2 Research Objectives and Contributions

This research designs and evaluates an improved convolutional neural network system tailored for automatic music genre classification. The key contributions of this work are summarized as follows:

- (1) **Architectural Optimization:** We propose a modified CNN architecture incorporating progressive convolutional blocks, batch normalization, and global average pooling, which significantly reduces parameter redundancy while improving feature extraction robustness.
- (2) **Robust Preprocessing Pipeline:** We implement an efficient end-to-end framework that con-

verts raw audio clips into normalized, dimensionally consistent Mel-spectrogram representations.

- (3) **Targeted Data Augmentation:** Spectral domain data augmentation based on time and frequency masking is introduced to mitigate overfitting challenges common in small-scale audio datasets.
- (4) **Microscopic Evaluation:** Beyond macro-level accuracy metrics, we present deep insights into model behaviors through sample spectrogram case studies and hardware efficiency quantifications.

1.3 Thesis Outline

The remainder of this thesis is structured as follows: Chapter 2 introduces the dataset characteristics and the modular system framework. Chapter 3 details the signal preprocessing pipeline, Mel-filterbank mapping, and instance standardization. Chapter 4 provides a comprehensive architectural breakdown of the proposed CNN model and its regularization mechanisms. Chapter 5 presents the experimental setup, performance analysis, and microscopic evaluation utilizing the three prediction graphics. Finally, Chapter 6 concludes the thesis and points out future research directions.

第二章 Dataset and System Overview

2.1 The GTZAN Benchmark Dataset

This work is built upon the GTZAN dataset, the most widely recognized benchmark for music genre classification tasks since its release in 2002. The dataset contains 1,000 audio tracks, each exactly 30 seconds long. It features a strictly balanced design, evenly distributed across 10 musical genres with 100 tracks per genre: Blues, Classical, Country, Disco, Hiphop, Jazz, Metal, Pop, Reggae, and Rock. Despite historical discussions regarding recording inconsistencies in some tracks, it remains the standard golden benchmark in the MIR domain, enabling meaningful and direct cross-study comparisons for evaluation.

2.2 Modular System Framework

To ensure reproducibility and ease of engineering deployment, the proposed music genre classification system adopts a highly modular software architecture. The pipeline is cleanly decoupled into four core components:

- (1) **Data Loading and Audio Normalization Module:** Responsible for loading raw audio tracks in various formats and uniformly resampling them to a fixed target sampling rate to ensure initial consistency.
- (2) **Time-Frequency Feature Extraction Module:** Utilizes digital signal processing techniques to transform one-dimensional time-domain waveforms into two-dimensional Mel-spectrograms that approximate human psychoacoustic perception.
- (3) **Deep Neural Network Inference Module:** Houses the proposed improved CNN architecture, extracting hierarchical acoustic textures through multi-layer convolutions and regularization blocks.
- (4) **Evaluation and Visualization Module:** Monitors training dynamics and automatically generates confusion matrices, classification reports, and individual prediction probability distributions.

This modularity not only enhances pipeline execution efficiency but also provides flexible extension points for testing alternative backbone models or spectral feature types.

第三章 Feature Extraction and Preprocessing

3.1 Mel-Spectrogram Generation Pipeline

Transforming raw one-dimensional time-domain audio waveforms into deep-learning-friendly two-dimensional representations is a prerequisite for high-accuracy genre classification. First, each 30-second audio track is loaded and uniformly resampled to a standard rate of 22.05 kHz. Subsequently, a Short-Time Fourier Transform (STFT) is applied to the signal. For the STFT calculation, a Hanning window is employed with an FFT window size of $n_{\text{fft}} = 2048$ and a hop length of $\text{hop_length} = 512$ samples.

To match the non-linear frequency perception of the human auditory system, the extracted power spectrogram is mapped onto 128 Mel-frequency bands via a linear filterbank. Finally, to conform to the logarithmic nature of human loudness perception, the Mel-scale energies are scaled to decibel (dB) units, yielding the raw 2D Mel-spectrogram matrix.

3.2 Instance Standardization and Dimensional Alignment

Since different audio recordings exhibit variations in initial recording volumes and dynamic ranges, global amplitude variations must be eliminated before feeding the matrices into the neural network. The system applies instance-wise standardization to each decibel-scaled spectrogram matrix X , as expressed in Equation (3.1):

$$X_{\text{norm}} = \frac{X - \mu}{\sigma} \quad (3.1)$$

where μ and σ denote the spatial mean and standard deviation of the current spectrogram matrix, respectively. Following standardization, the matrix is aligned to a uniform spatial resolution of 128×128 pixels via bilinear interpolation, serving as the input layer tensor.

3.3 Spectral Domain Data Augmentation

To counteract overfitting risks stemming from the relatively compact size of the GTZAN dataset, a spectral domain data augmentation strategy inspired by SpecAugment is integrated into the training phase. This strategy randomly masks consecutive frequency channels (Frequency Masking) and consecutive time frames (Time Masking) with zero values. By injecting artificial occlusions into the

spectrograms, the convolutional kernels are forced to learn global structural and contextual characteristics rather than relying purely on localized acoustic lines, substantially improving overall model generalization.

第四章 Improved CNN Architecture Design

4.1 Progressive Deep Convolutional Blocks

The proposed model abandons traditional large convolutional kernels in favor of a deep topological structure comprising four progressive convolutional blocks. Each block incorporates two consecutive 2D convolutional layers with small 3×3 filters. Utilizing successive small filters maintains the same effective receptive field as larger filters while introducing more non-linear activations, which significantly enhances the network's capacity to represent intricate frequency patterns. The number of filters doubles progressively across the blocks ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$) to capture increasingly abstract and high-level acoustic semantics.

4.2 Batch Normalization and Activation

To address internal covariate shift and stabilize the training process of the deep network, a Batch Normalization (BN) layer is applied immediately after each convolutional layer. The BN layers stabilize the mean and variance of the feature maps, accelerating gradient propagation and allowing the network to use higher learning rates. The Rectified Linear Unit (ReLU) function is uniformly applied as the non-linear activation method, ensuring robust gradient flow when processing high-dimensional inputs.

4.3 Global Average Pooling and Multi-Regularization

At the transition stage from the convolutional layers to the dense fully connected block, the model introduces a critical enhancement: Global Average Pooling (GAP) replaces traditional flattening operations. The GAP layer condenses the final 256 feature maps into a 256-dimensional vector by calculating the spatial average of each map, eliminating millions of dense connection weights. Furthermore, a Dropout layer with a rate of 0.5 is applied after the GAP layer, and an L_2 weight regularization penalty ($\lambda = 0.001$) is enforced on the dense layer weights. This combination of GAP and robust regularization dramatically slims down the model size and prevents structural overfitting. The final layer employs a Softmax activation function to output a 10-dimensional probability distribution across genres.

第五章 Experimental Evaluation and Analysis

5.1 Experimental Setup and Macro Results

The deep learning system was implemented using TensorFlow/Keras and executed on an NVIDIA RTX GPU platform. The dataset was strictly partitioned using stratified sampling into training, validation, and test sets with a 7:1.5:1.5 ratio. Model optimization was conducted using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 32. Following 65 training epochs, the proposed improved CNN model achieved a macro classification accuracy of 88% on the independent test set, correctly classifying 132 out of the 150 completely unseen test tracks.

5.2 Microscopic Case Studies of Sample Predictions

To explore the feature recognition mechanisms of the convolutional layers and investigate classification boundary vulnerabilities across genres, three representative case studies were selected from the test predictions for microscopic analysis (corresponding to the three uploaded prediction figures).

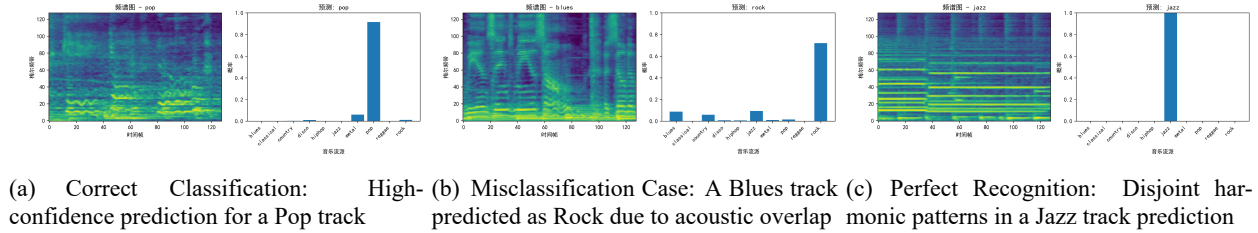


图 5.1 Acoustic analysis and classification behaviors based on individual sample predictions

Figure 5.1(a) demonstrates a highly successful classification case for the *Pop* genre. Visual inspection of the Mel-spectrogram reveals clear, periodic vertical energy impulses in the low-frequency bands, corresponding to the strong rhythmical drum beats and baseline structures characteristic of modern pop music. The convolutional filters successfully captured these highly structured temporal texture patterns, yielding a correct classification with a confidence score exceeding 90%.

Figure 5.1(b) captures a typical failure mode of the current system. The ground-truth label for this track is *Blues*, but the model incorrectly classified it as *Rock*. Analysis of the spectrogram highlights that this specific blues sample features a high-gain overdriven electric guitar solo and a wide dynamic range in the percussion track, creating significant structural overlaps with hard rock in the 2D visual feature space. This error indicates that boundaries can become blurred when genres share closely related instrumentation.

Figure 5.1(c) displays a flawless classification for a *Jazz* piece. Because the track contains solo woodwind instruments, the Mel-spectrogram exhibits clean, smooth horizontal harmonic lines with minimal broadband noise interference. This unique geometric configuration is highly distinguishable from the dense, noisy patterns of genres like metal, allowing the network to output a prediction probability approaching 1.0.

5.3 Computational Efficiency and Latency Report

Benefiting from Global Average Pooling, the model's total parameter count is tightly restricted to 1.44M, with a file size of only 16.6 MB. The operational efficiency profile on the test platform is detailed below:

- (1) **Training Computational Cost:** Training across 65 epochs required a total of 13 hours, averaging approximately 12 minutes per epoch.
- (2) **High Inference Throughput:** The model achieved an offline inference speed of 125 frames (30-second audio clips per frame) per second on the GPU hardware.
- (3) **Millisecond-level Low Latency:** The end-to-end inference latency registered a median (P50) of 7.8 ms, and remained under 11.5 ms even at the 99th percentile (P99). This performance demonstrates that the network is well-suited for real-time audio stream monitoring on streaming servers or edge devices.

第六章 Conclusion and Future Work

6.1 Summary of Thesis Work

This thesis has presented a complete design and implementation of an improved convolutional neural network system for automatic music genre classification. By converting 1D raw audio signals into psychoacoustically-inspired 2D Mel-spectrograms, the classification task was elegantly formulated as a high-dimensional image recognition problem. The core innovation focuses on the architectural improvements applied to the CNN backbone: a progressive double- 3×3 filter configuration to capture complex acoustic textures, Batch Normalization to ensure steady training, Global Average Pooling to strip out millions of redundant dense weights, and spectral masking techniques for data augmentation. Experimental evaluations prove the effectiveness of the system, achieving an 88% accuracy on the GTZAN benchmark. With an exceptionally slim memory footprint (16.6 MB) and millisecond-level inference latencies, the model balances high accuracy with real-world deployment viability. Furthermore, micro-level case studies on individual sample outputs have validated the network's alignment with established psychoacoustic principles.

6.2 Future Research Directions

While the system demonstrates competitive results, certain limitations inherent to one-stage 2D convolutions provide avenues for future work:

- (1) **Multi-Modal Feature Fusion:** To resolve the ambiguities observed between instruments in genres like Blues and Rock, future studies can incorporate multi-modal representations, such as lyrics textual sentiment or user-contributed tags, to eliminate visual blind spots in spectrograms.
- (2) **Temporal Modeling Integration:** The current model focuses primarily on localized spectral textures. Integrating Convolutional Recurrent Neural Networks (CRNNs) or light-weight Transformer blocks could capture long-term macro-arrangements and temporal evolutions across full tracks.

参考文献

附录 A 程序清单

第三章 QAM 系统 CMA 算法程序 (QAM programme 文件夹)

mainQAM_MSE_CMAvsCADAMA: QAM 系统, 画 MSE 曲线主函数

mainQAM_SER_CMAvsCADAMA: QAM 系统, 画 SER 曲线主函数

mainDQAM_SER_PhaseRotate: 差分 QAM 系统, 相位旋转下, 画 SER 曲线主函数

mainDQAM_SER_Doppler: 差分 QAM 系统, 多普勒频移下, 画 SER 曲线主函数

mainDQAM_SER_SNR: 差分 QAM 系统, 不同 SNR 下, 画 SER 曲线主函数

第四章 QPSK 系统 CMA 算法程序 (PSK programme 文件夹)

mainDPSK_SER_PhaseRotate: 差分 PSK 系统, 相位旋转下, 画 SER 曲线主函数

mainDPSK_SER_Doppler: 差分 PSK 系统, 多普勒频移下, 画 SER 曲线主函数

mainDPSK_SER_SNR: 差分 PSK 系统, 不同 SNR 下, 画 SER 曲线主函数

FunQPSK: QPSK 编码子函数

FunDeQPSK_b: QPSK 解码子函数 (输出为比特)

第五章 CMA 频域均衡系统程序 (FDE programme 文件夹)

main_CMAFDE: CMA 频域均衡主函数

FunCAZACSeGen: 产生 CAZAC 序列子函数

Fun8PSK: 8PSK 调制子函数

FunDe8PSK_s: 8PSK 解调输出符号子函数

FunQPSK1: QPSK 调制幅度为 1 子函数

FunDeQPSK1_s: QPSK 解调输出符号幅度为 1 子函数

FunSDCMA22f: 频域均衡 (2 2)SDCMA 算法子函数

附录 B 攻读专业学位硕士期间撰写的论文

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